Thermal gyroresonance emission in radiosynth

Equations, derivation and numerical implementation

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radiosynth 2.2.0 — *From MHD to MHz* — November 2026

https://github.com/soumitrahazra/radiosynth

Abstract

This note collects every equation used by radiosynth to compute the thermal gyroresonance (GR) emission of a magnetohydrodynamic (MHD) model, together with the magnetised free–free emission that accompanies it, in Stokes I and V. It starts from the cold-plasma magneto-ionic modes, derives the optical depth of a harmonic layer from the low-temperature expansion of the gyromagnetic absorption coefficient, and then describes how the code finds the layers along straight lines of sight, how it solves the polarised radiative transfer for the two circular polarisations (or the two modes), and how it turns the result into brightness temperatures, polarisation and fluxes. Each section ends with the corresponding function of the code (`radiosynth/gyroresonance/`, `common/magnetoionic.py`, `bremsstrahlung/magnetized.py`), so that the equations can be checked line by line.

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1 Overview

A thermal electron gyrating in a magnetic field B radiates and absorbs at the harmonics $s\nu_B$ of its gyrofrequency $\nu_B = eB/(2\pi m_e)$. In a Maxwellian plasma with $k_\mathrm{B}T_e \ll m_e c^2$ the absorption at a given observing frequency ν is therefore confined to thin *harmonic layers*, the surfaces on which

$$\nu = s\,\nu_B(\boldsymbol{r}), \qquad s = 1, 2, 3, \ldots \tag{1}$$

Over active regions, where B reaches several hundred to a few thousand gauss, the layers with $s = 2$ and $s = 3$ are usually optically thick at 1–20 GHz, so the brightness temperature of the radio source equals the coronal electron temperature at the height of the layer. Gyroresonance emission is the main microwave signature of sunspots, and its circular polarisation measures the coronal magnetic field.

radiosynth computes this emission in five steps (Fig. 1):

1. set up the image plane and one straight line of sight per pixel, and sample n_e, T_e and $\boldsymbol{B}$ of the model at equally spaced nodes along it (Sect. 3);
2. evaluate the magneto-ionic parameters and the mode polarisations (Sect. 4);
3. compute the magnetised free–free optical depth of every step (Sect. 7) and the optical depth of every harmonic layer crossed in the step (Sects. 5–8);
4. solve the radiative transfer for the two polarisations (Sects. 6, 9 and 10);
5. convert to Stokes I, V, brightness temperatures and fluxes, and compute the diagnostics (Sects. 11 and 12).

Refraction is negligible where gyroresonance emission forms ($\nu \gg \nu_p$), so the rays are straight; the refractive ray tracer of radiosynth is used only for the free–free and simplistic mechanisms.

2 Notation, units and constants

The code works in SI units, with the magnetic field in gauss and lengths of the model in coordinate units (solar or stellar radii) converted to metres with the grid's `length_unit`. In SI, e^2 is replaced by $e^2/4\pi\varepsilon_0$ in every Gaussian formula below. The main quantities are:

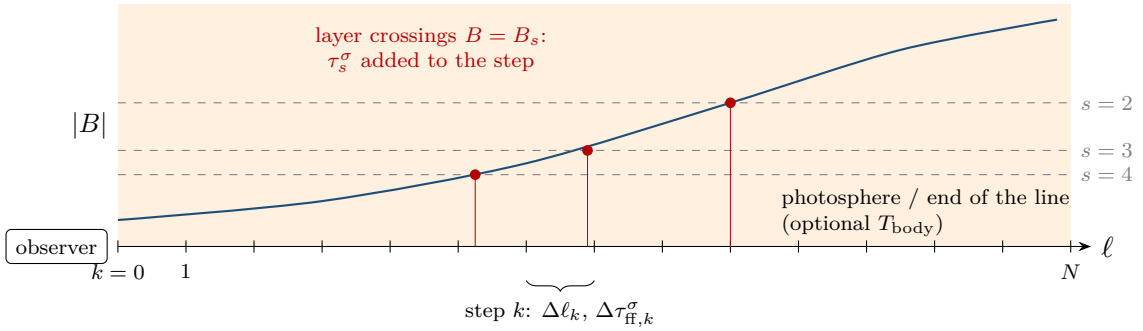


Figure 1: How a line of sight is treated. The model is sampled at nodes $k = 0 \ldots N$, from the observer inwards. Each step carries the free–free optical depth of its mid-point, plus the optical depth of every harmonic layer $\nu = s\nu_B$ whose field strength lies between the field strengths of its two nodes. The layers are much thinner than a step and are treated as discontinuities.

Symbol	Meaning	Value / unit
ν	observing frequency	Hz
n_e, T_e	electron density, electron temperature	m^{-3}, K
$\boldsymbol{B}$, $B = \lvert\boldsymbol{B}\rvert$	magnetic field and its strength	G
$\hat{\boldsymbol{k}}$	unit vector along the line of sight, *towards the observer*	—
θ	angle between $\boldsymbol{B}$ and $\hat{\boldsymbol{k}}$, $\cos\theta = \boldsymbol{B}\cdot\hat{\boldsymbol{k}}/B$	—
ν_B	electron gyrofrequency, $eB/(2\pi m_e)$	$2.799249\,\mathrm{MHz\,G^{-1}} \times B$
ν_p	plasma frequency, $\nu_p^2 = e^2 n_e/(4\pi^2\varepsilon_0 m_e)$	Hz
$n_{\rm cr}$	critical density, $n_{\rm cr} = \pi m_e \nu^2/(e^2/4\pi\varepsilon_0)$	$1.2404\times10^{16}\,\mathrm{m^{-3}}$ at 1 GHz
X	$(\nu_p/\nu)^2 = n_e/n_{\rm cr}$	—
Y	ν_B/ν ($Y = 1/s$ on the s-th layer)	—
β_T^2	$k_{\rm B}T_e/(m_e c^2)$	1.6864×10^{-4} at 1 MK
L_B	magnetic scale length along the ray, $\lvert B/(\mathrm{d}B/\mathrm{d}\ell)\rvert$	m
σ	magneto-ionic mode: x (extraordinary) or o (ordinary)	—
T_σ	axial ratio of the polarisation ellipse of mode σ	—
s	harmonic number, $1 \le s \le s_{\max}$ (default $s_{\max} = 10$)	—
$\pi e^2/(m_e c)$	prefactor of the layer optical depth	$2.6540\times10^{-6}\,\mathrm{m^2\,s^{-1}}$ ($0.026540\,\mathrm{cm^2\,s^{-1}}$ in cgs)
S_σ	source function of one mode, $\nu^2 k_{\rm B}T_e/c^2$	$\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$

In the code. `common/constants.py`: `E2_JM` $= e^2/4\pi\varepsilon_0$, `critical_density`; `common/magnetoionic.py`: `NU_B_PER_GAUSS`, `magnetoionic_params(ne, b, freq)` returns (X, Y); `gyroresonance/opacity.py`: `GR_PREFACTOR` $= \pi e^2/(m_e c)$.

3 Lines of sight and sampling

The image plane, the observer and the pixel grid are the same as for the free–free images of radiosynth. For pixel (i, j) the line of sight starts at the point $\boldsymbol{r}_0$ where it enters both the model domain and the integration sphere, and runs along the unit vector $\hat{\boldsymbol{d}}$ pointing *away* from the observer, so that

$$\hat{\boldsymbol{k}} = -\hat{\boldsymbol{d}}. \tag{2}$$

It ends where it leaves the domain or the integration sphere, or where it reaches the inner boundary $r = r_{\rm body}$, at the parameter $t_{\rm end}$ (coordinate units). The line is cut into steps of length Δs,

$$t_k = \min(k\,\Delta s,\ t_{\rm end}), \qquad \boldsymbol{r}_k = \boldsymbol{r}_0 + t_k\hat{\boldsymbol{d}}, \qquad k = 0, 1, \ldots, N, \tag{3}$$

with node 0 closest to the observer, and

$$\Delta\ell_k = (t_{k+1} - t_k)\, L_{\rm unit} \quad [\mathrm{m}], \tag{4}$$

where $L_{\rm unit}$ is the length unit of the grid ($R_\odot$ or R_*). By default Δs is half the smallest cell of the grid, and never less than $t_{\rm end}/20000$ for the longest line. At every node the model is interpolated to give $n_{e,k}$, $T_{e,k}$ and $\boldsymbol{B}_k$ (set to zero outside the domain), and

$$B_k = |\boldsymbol{B}_k|, \qquad \cos\theta_k = \frac{\boldsymbol{B}_k \cdot \hat{\boldsymbol{k}}}{B_k}\ (0 \text{ if } B_k = 0), \qquad X_k = \frac{n_{e,k}}{n_{\rm cr}}, \qquad Y_k = \frac{\nu_{B,k}}{\nu}. \tag{5}$$

A step is used only if both of its nodes lie inside the domain and $\Delta\ell_k > 0$.

Because the layers are located *between* the nodes (Sect. 8), Δs only has to resolve the variation of n_e, T_e and $\boldsymbol{B}$, not the layers themselves.

In the code. `common/straight.py`: `segment_ends` ($t_{\rm end}$ and whether the line ends on the body), `default_step` (Δs), `iter_samples` (nodes, n_e, T_e, $\boldsymbol{B}$, $\Delta\ell$, in chunks of rays). The keyword `ds` of `make_gr_image` sets Δs.

4 Magneto-ionic modes

4.1 Refractive indices and cutoffs

A cold, collisionless, magnetised electron plasma supports two transverse electromagnetic modes: the *extraordinary* (x) mode, whose electric vector rotates in the same sense as the gyrating electrons, and the *ordinary* (o) mode. Their refractive indices are given by the Appleton–Hartree formula

$$n_\sigma^2 = 1 - \frac{X(1-X)}{(1-X) - \frac{1}{2}Y^2\sin^2\theta \pm \sqrt{\frac{1}{4}Y^4\sin^4\theta + (1-X)^2Y^2\cos^2\theta}}, \tag{6}$$

with the upper sign for the o-mode and the lower sign for the x-mode. A mode cannot propagate where $n_\sigma^2 \le 0$; the cutoffs are

$$\text{o-mode: } X \ge 1, \qquad\qquad \text{x-mode: } X \ge 1 - Y \quad (Y < 1). \tag{7}$$

On the fundamental layer ($s = 1$, $Y = 1$) the x-mode is therefore cut off for any $X > 0$.

4.2 Polarisation of the modes

In the frame with $\boldsymbol{B}$ along $\hat{z}$ and $\hat{\boldsymbol{k}}$ in the x–z plane, the transverse electric vector of mode σ is (longitudinal part neglected)

$$\boldsymbol{e}_\sigma = \frac{(T_\sigma\cos\theta,\ i,\ -T_\sigma\sin\theta)}{\sqrt{1+T_\sigma^2}}, \qquad T_oT_x = -1, \tag{8}$$

where T_σ is the axial ratio of the polarisation ellipse. From the Appleton–Hartree theory,

$$|T_x| = \frac{2(1-X)|\cos\theta|}{Y\sin^2\theta + \sqrt{Y^2\sin^4\theta + 4(1-X)^2\cos^2\theta}}, \qquad |T_o| = \frac{1}{|T_x|}. \tag{9}$$

The limits are

- quasi-longitudinal propagation ($Y\sin^2\theta \ll 2(1-X)|\cos\theta|$): $|T_x| \to 1$, both modes circularly polarised;
- quasi-transverse propagation ($\theta \to 90°$): $T_x \to 0$, the x-mode is linearly polarised with $\boldsymbol{E} \perp \boldsymbol{B}$ and the o-mode with $\boldsymbol{E} \parallel \boldsymbol{B}$.

4.3 Handedness and Stokes V

radiosynth uses the IAU/IEEE convention: Stokes $V > 0$ for right-hand circular polarisation, i.e. an electric vector rotating counter-clockwise when seen by the observer (looking towards the source). For $\boldsymbol{B}\cdot\hat{\boldsymbol{k}} > 0$ (field pointing towards the observer) the x-mode is right-hand polarised and contributes $V > 0$; for $\boldsymbol{B}\cdot\hat{\boldsymbol{k}} < 0$ it is left-hand polarised. Gyroresonance emission, dominated by the x-mode, is therefore right-hand polarised above a positive-polarity spot seen from above.

In the code. `common/magnetoionic.py`: `refractive_index2` (Eq. 6), `cutoff` (Eq. 7), `axial_ratio_x` (Eq. 9; $1-X$ is floored at 10^{-12}, and $|T_x| = 1$ where the denominator vanishes).

5 Gyroresonance opacity of a harmonic layer

5.1 Resonance and line width

An electron with velocity components $\beta_\parallel c$ along and $\beta_\perp c$ across $\boldsymbol{B}$ interacts with a wave of frequency ν and refractive index n_σ at the Doppler-shifted resonance

$$\nu\left(1 - n_\sigma \beta_\parallel \cos\theta\right) = \frac{s\,\nu_B}{\gamma}. \tag{10}$$

For a thermal plasma ($\gamma \simeq 1$) the absorption at harmonic s is a narrow line around $\nu = s\nu_B$, broadened by the thermal motion along $\boldsymbol{B}$. With $n_\sigma \simeq 1$ its profile is Gaussian,

$$\phi_s(\nu) = \frac{1}{\sqrt{\pi}\,\Delta\nu_D}\exp\left[-\frac{(\nu - s\nu_B)^2}{\Delta\nu_D^2}\right], \qquad \Delta\nu_D = \sqrt{2}\,\nu\,\beta_T|\cos\theta|, \qquad \int \phi_s\,\mathrm{d}\nu = 1. \tag{11}$$

(For $\cos\theta \to 0$ the width is set instead by the relativistic mass change, of order $\nu\beta_T^2$; the layer is then even thinner.)

5.2 Strength of the line: low-temperature expansion

The coupling of an electron to mode σ at harmonic s contains the Bessel functions $J_{s\pm1}(x)$ and $J_s(x)$ of argument

$$x = s\,\beta_\perp \sin\theta \tag{12}$$

(Schott's formula; e.g. Melrose 1980). In the corona $\beta_T^2 \approx (1.7\text{–}5)\times10^{-4}$, so $x \ll 1$ and only the lowest-order term survives. For the component of the wave field that co-rotates with the electrons the dominant term is

$$J_{s-1}(x) \simeq \frac{(x/2)^{s-1}}{(s-1)!}. \tag{13}$$

The perpendicular velocities of a Maxwellian are distributed as

$$f(\beta_\perp)\,\mathrm{d}\beta_\perp \propto \exp\left[-\frac{\beta_\perp^2}{2\beta_T^2}\right]\beta_\perp\,\mathrm{d}\beta_\perp, \tag{14}$$

whose moments are

$$\left\langle \beta_\perp^{2m}\right\rangle = m!\,\left(2\beta_T^2\right)^m. \tag{15}$$

Hence

$$\begin{aligned} s\left\langle J_{s-1}^2(s\beta_\perp\sin\theta)\right\rangle &\simeq s\,\frac{(s\sin\theta/2)^{2(s-1)}}{[(s-1)!]^2}\left\langle\beta_\perp^{2(s-1)}\right\rangle = s\,\frac{(s^2\sin^2\theta/4)^{s-1}\,(s-1)!\,(2\beta_T^2)^{s-1}}{[(s-1)!]^2} \\ &= \frac{s^2}{s!}\left(\frac{s^2\beta_T^2\sin^2\theta}{2}\right)^{s-1}\left[1 + O(s^2\beta_T^2)\right]. \end{aligned} \tag{16}$$

This factor carries the very steep dependence of the opacity on s, T_e and θ: each additional harmonic costs a factor $\sim s^2\beta_T^2\sin^2\theta/2 \sim 10^{-3}$.

The projection of the wave polarisation $\boldsymbol{e}_\sigma$ onto the co-rotating component gives the polarisation factor

$$F_\sigma(\theta) = \frac{(1 + T_\sigma \cos\theta)^2}{1 + T_\sigma^2}, \tag{17}$$

with T_σ from Eq. (9) evaluated on the layer ($Y = 1/s$) and signed so that $T_x \cos\theta > 0$ and $T_o = -1/T_x$. Explicitly,

$$F_x = \frac{(1 + |T_x||\cos\theta|)^2}{1 + T_x^2}, \qquad F_o = \frac{(|T_x| - |\cos\theta|)^2}{1 + T_x^2}. \tag{18}$$

Along $\boldsymbol{B}$: $F_x = 2$, $F_o = 0$; across $\boldsymbol{B}$: $F_x = 1$, $F_o = 0$ (Fig. 2, right). The x-mode is always much more strongly absorbed than the o-mode.

5.3 Frequency-integrated absorption coefficient

Combining these factors, the absorption coefficient of mode σ near harmonic s is a line of fixed area,

$$\kappa_s^\sigma(\nu, \ell) = \mathcal{A}_s^\sigma \, \phi_s(\nu - s\nu_B(\ell)), \qquad \mathcal{A}_s^\sigma = \frac{\pi e^2}{m_e c} \, n_e \, \frac{s^2}{s!} \left(\frac{s^2\beta_T^2 \sin^2\theta}{2}\right)^{s-1} F_\sigma(\theta), \tag{19}$$

to lowest order in β_T and with $n_\sigma = 1$ (the full expression contains a factor n_σ^{2s-3}; Zheleznyakov 1970; Melrose 1980; White & Kundu 1997; Fleishman & Kuznetsov 2014). $\mathcal{A}_s^\sigma$ has units of $\mathrm{m^{-1}\,Hz}$.

5.4 Integration across the layer

At fixed observing frequency, ν_B varies along the ray, so the line (11) is crossed in a layer of thickness

$$\delta\ell \simeq L_B \, \frac{\Delta\nu_D}{\nu} = \sqrt{2}\,\beta_T |\cos\theta| \, L_B, \qquad L_B = \left|\frac{B}{\mathrm{d}B/\mathrm{d}\ell}\right|, \tag{20}$$

about 1–2 % of L_B in the corona. Over this thickness n_e, T_e, θ and $\mathrm{d}\nu_B/\mathrm{d}\ell$ are constant, and the optical depth is

$$\tau_s^\sigma = \int \kappa_s^\sigma \, \mathrm{d}\ell = \mathcal{A}_s^\sigma \int \phi_s(\nu - s\nu_B(\ell)) \, \mathrm{d}\ell = \frac{\mathcal{A}_s^\sigma}{s\,|\mathrm{d}\nu_B/\mathrm{d}\ell|} \int \phi_s(\nu - u) \, \mathrm{d}u = \frac{\mathcal{A}_s^\sigma}{s\,|\mathrm{d}\nu_B/\mathrm{d}\ell|}, \tag{21}$$

with $u = s\nu_B(\ell)$. On the layer $s\nu_B = \nu$ and $|\mathrm{d}\nu_B/\mathrm{d}\ell| = \nu_B/L_B$, so $s|\mathrm{d}\nu_B/\mathrm{d}\ell| = \nu/L_B$. The result does not depend on the line profile:

$$\boxed{\tau_s^\sigma = \frac{\pi e^2}{m_e c} \, \frac{n_e \, L_B}{\nu} \, \frac{s^2}{s!} \left(\frac{s^2\beta_T^2 \sin^2\theta}{2}\right)^{s-1} F_\sigma(\theta)} \tag{22}$$

with n_e, T_e and θ taken on the layer, $\nu = s\nu_B$.

5.5 Orders of magnitude

Table 2 gives Eq. (22), as evaluated by the code, for typical active-region values. The $s = 2$ layer is very thick in the x-mode and usually also in the o-mode, $s = 3$ is thick in the x-mode and marginal in the o-mode, and $s \geq 4$ is thin. This is why the microwave brightness of a sunspot at frequency ν is set by the temperature where $B = \nu/(3\nu_B/\mathrm{G})$ (x-mode) and $B = \nu/(2\nu_B/\mathrm{G})$

Table 2: Layer optical depths, Eq. (22), computed with `layer_optical_depth` for $n_e = 10^9\,\mathrm{cm}^{-3}$, $T_e = 2\,\mathrm{MK}$, $L_B = 10^9\,\mathrm{cm}$ at $\nu = 5\,\mathrm{GHz}$ ($X = 3.2 \times 10^{-3}$). At $s = 1$ the x-mode is cut off in the transfer (Sect. 8).

	$\theta = 30°$		$\theta = 60°$		$\theta = 80°$	
s	τ^x	τ^o	τ^x	τ^o	τ^x	τ^o
1	(9.3×10^6)	0.49	(6.6×10^6)	4.0	(5.5×10^6)	1.4
2	3.1×10^3	3.9	6.6×10^3	134	7.0×10^3	135
3	2.0	4.5×10^{-3}	12.4	0.50	16.8	0.99
4	1.9×10^{-3}	5.5×10^{-6}	0.035	1.9×10^{-3}	0.059	6.0×10^{-3}
5	2.4×10^{-6}	7.8×10^{-9}	1.3×10^{-4}	8.3×10^{-6}	2.8×10^{-4}	3.9×10^{-5}
6	3.7×10^{-9}	1.3×10^{-11}	6.1×10^{-7}	4.3×10^{-8}	1.7×10^{-6}	2.9×10^{-7}

Table 3: Field strength [G] of the harmonic layers, $B_s = \nu/(s \times 2.799249\,\mathrm{MHz\,G^{-1}})$ (`harmonic_field`).

ν	$s = 1$	$s = 2$	$s = 3$	$s = 4$
1 GHz	357	179	119	89
2 GHz	714	357	238	179
5 GHz	1786	893	595	447
10 GHz	3572	1786	1191	893
17 GHz	6073	3037	2024	1518

(o-mode), and why its emission is strongly x-mode (right-hand over positive field) polarised where the o-mode is still thin. Table 3 lists the field strengths of the layers.

In the code. `gyroresonance/opacity.py`: `mode_factor(X, Y, cos, mode)` returns F_σ (Eq. 17) with $|T_x|$ from `axial_ratio_x` at $Y = 1/s$; `layer_optical_depth(s, ne, te, cos, l_b, freq, mode, X)` returns Eq. (22) with $\beta_T^2 = k_\mathrm{B}T_e/(m_e c^2)$ and $\sin^2\theta = \max(1 - \cos^2\theta, 0)$; `harmonic_field(s, freq)` returns B_s.

6 Kirchhoff's law and the source function

For a thermal plasma Kirchhoff's law holds for each mode separately. With $n_\sigma = 1$ each mode carries half of the Rayleigh–Jeans intensity, so the source function of mode σ is

$$S_\sigma = \frac{\nu^2 k_\mathrm{B} T_e}{c^2} = \frac{1}{2} B_\nu^\mathrm{RJ}(T_e). \tag{23}$$

An optically thick layer therefore has brightness temperature T_e *in that mode*, and an optically thick isothermal source that is thick in both modes is unpolarised.

In the code. `gyroresonance/transfer.py`: the constant `_SRC` $= k_\mathrm{B}/c^2$, so that $S =$ `_SRC` $\nu^2 T$; `bremsstrahlung/magnetized.py`: `rayleigh_jeans_mode`.

7 Magnetised free–free absorption

7.1 Unmagnetised coefficient

The thermal free–free (bremsstrahlung) absorption coefficient is, in the default of the gyroresonance images, the Coulomb-logarithm form of Dulk (1985, Eq. 21),

$$\kappa_\mathrm{ff} = 9.78 \times 10^{-3}\, \frac{n_e^2}{\nu^2 T_e^{3/2}}\, G \quad [\mathrm{cm}^{-1},\ n_e \text{ in } \mathrm{cm}^{-3}], \tag{24}$$

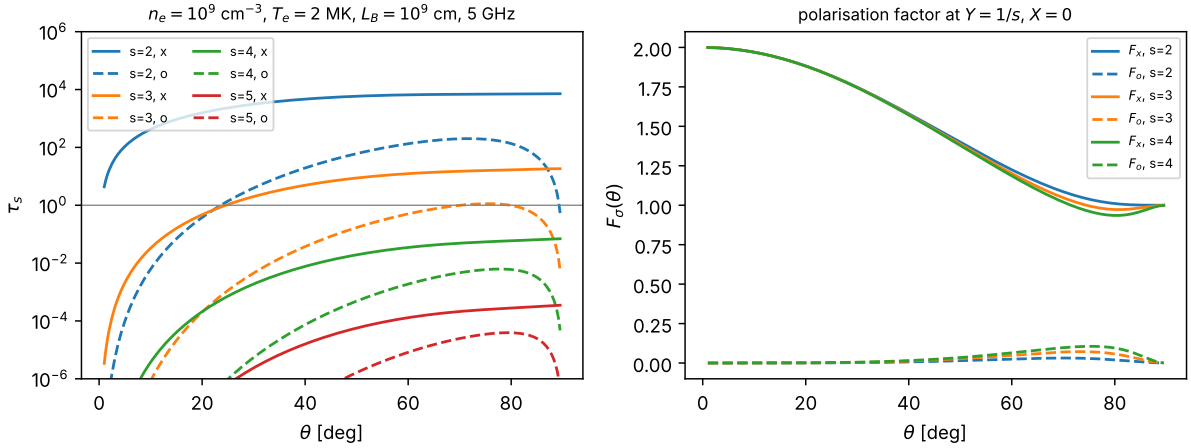


Figure 2: Left: layer optical depth, Eq. (22), against the angle between $\boldsymbol{B}$ and the line of sight for $s = 2$–5 (solid: x-mode, dashed: o-mode), with the parameters of Table 2. Right: the polarisation factor $F_\sigma(\theta)$, Eq. (17), on the layers $s = 2$, 3, 4 ($X = 0$). Both panels were computed with the functions of `gyroresonance/opacity.py`.

$$G = \begin{cases} 18.2 + \ln T_e^{3/2} - \ln \nu, & T_e < 2 \times 10^5\,\mathrm{K}, \\ 24.5 + \ln T_e - \ln \nu, & T_e \geq 2 \times 10^5\,\mathrm{K}, \end{cases} \tag{25}$$

with $G \geq 1$, converted to m^{-1}. Alternatively (`gaunt`$= g$, a number) the code uses the expression of Moschou et al. (2018) with a constant Gaunt factor,

$$\kappa_{\mathrm{ff}} = \frac{4}{3}\sqrt{\frac{2\pi}{3}}\left[\frac{e^2/4\pi\varepsilon_0}{\sqrt{k_\mathrm{B} T_e\, m_e}}\right]^3 \left(\frac{n_e}{\nu}\right)^2 \frac{g}{c}. \tag{26}$$

Both use the total source function $B_\nu^{\mathrm{RJ}} = 2\nu^2 k_\mathrm{B} T_e / c^2$. For the parameters of Table 2, $\kappa_{\mathrm{ff}} = 2.3 \times 10^{-10}\,\mathrm{m}^{-1}$ (Dulk) or $2.5 \times 10^{-10}\,\mathrm{m}^{-1}$ ($g = 10$): over 10^9 cm the corona is thin in free–free at 5 GHz, while the $s = 2$ and 3 layers are thick.

7.2 The two modes

In a magnetic field the free–free absorption differs between the modes. For $Y < 1$ and away from quasi-transverse propagation (Zheleznyakov 1970; Dulk 1985; Gelfreikh 2004),

$$\kappa_\sigma = \frac{\kappa_{\mathrm{ff}}}{(1 + \sigma q)^2}, \qquad q = \min(Y|\cos\theta|,\ 0.9)\,, \qquad \sigma = +1\ (\mathrm{o}),\ -1\ (\mathrm{x}), \tag{27}$$

with S_σ from Eq. (23) for each mode, so that the two modes together give B_ν^{RJ} in the thick limit. The cap $q \leq 0.9$ avoids the divergence of the approximation near the fundamental layer ($Y \to 1$).

For an optically thin source the degree of circular polarisation is

$$\frac{|V|}{I} = \frac{\kappa_x - \kappa_o}{\kappa_x + \kappa_o} = \frac{(1+q)^2 - (1-q)^2}{(1+q)^2 + (1-q)^2} = \frac{2q}{1+q^2} \simeq 2Y|\cos\theta|, \tag{28}$$

in the sense of the x-mode; for a thick isothermal source $V \to 0$.

In the code. `bremsstrahlung/emission.py`: `bremsstrahlung_coefficients` (Eq. 24 or the constant-g form; T_e is floored at 1 K); `bremsstrahlung/magnetized.py`: `magnetized_ff_coefficients` returns $(\kappa_o, \kappa_x, S_\sigma)$ of Eq. (27) (`y_max`$= 0.9$). Keyword `gaunt` of `make_gr_image` (default `"dulk"`).

8 Discretisation: steps and layers

8.1 Free–free part of a step

The free–free optical depth of step k is evaluated at its mid-point, with the arithmetic means $\bar{q}_k = \frac{1}{2}(q_k + q_{k+1})$ of n_e, T_e, $\cos\theta$ and B:

$$\Delta\tau^{\sigma}_{\mathrm{ff},k} = \kappa_\sigma\Big(\bar{n}_{e,k}, \bar{T}_{e,k}, \bar{B}_k, \overline{\cos\theta}_k\Big)\,\Delta\ell_k. \tag{29}$$

8.2 Finding the layers

Harmonic s has a layer in step k if $s\nu_B - \nu$ changes sign between the nodes,

$$(s\,\nu_{B,k} - \nu)\,(s\,\nu_{B,k+1} - \nu) < 0. \tag{30}$$

The crossing is located by linear interpolation of ν_B,

$$u = \frac{\nu/s - \nu_{B,k}}{\nu_{B,k+1} - \nu_{B,k}} \in (0,1), \tag{31}$$

and the plasma parameters on the layer are interpolated with the same weight,

$$q_c = q_k + u\,(q_{k+1} - q_k) \qquad \text{for } n_e,\ T_e,\ \cos\theta,\ X. \tag{32}$$

The magnetic scale length is the finite-difference estimate

$$L_B = \frac{B_s\,\Delta\ell_k}{|B_{k+1} - B_k|}, \qquad B_s = \frac{\nu}{s\,\nu_B/\mathrm{G}}, \tag{33}$$

i.e. $|B/(\mathrm{d}B/\mathrm{d}\ell)|$ on the layer. With these values Eq. (22) gives $\tau^{x}_{s,k}$ and $\tau^{o}_{s,k}$. On the fundamental layer the x-mode is cut off for any $X > 0$ (Eq. 7 with $Y = 1$), so

$$\tau^{x}_{1,k} = 0 \quad \text{if } X_c > 0. \tag{34}$$

All harmonics $1 \le s \le s_{\max}$ are tested in every step; several layers can fall into the same step. Because the layers are thinner than a step, treating each of them as a discontinuity of optical depth is exact as long as Δs resolves L_B; one harmonic crossed twice within one step (if B is not monotonic within it) would be missed, which is a further reason to keep Δs below the scale of the field.

8.3 Total optical depth and source function of a step

For each mode the step has

$$\Delta\tau^{\sigma}_k = \Delta\tau^{\sigma}_{\mathrm{ff},k} + \sum_s \tau^{\sigma}_{s,k}, \qquad W^{\sigma}_k = \Delta\tau^{\sigma}_{\mathrm{ff},k}\,\bar{T}_{e,k} + \sum_s \tau^{\sigma}_{s,k}\,T_{e,c}, \tag{35}$$

and its source function is the optical-depth-weighted mean

$$S^{\sigma}_k = \frac{\nu^2 k_{\mathrm{B}}}{c^2}\,\frac{W^{\sigma}_k}{\Delta\tau^{\sigma}_k} \qquad (0 \text{ if } \Delta\tau^{\sigma}_k = 0). \tag{36}$$

This is exact for an isothermal step and keeps Kirchhoff's law: a step that is thick only in one of its layers emits at the temperature of that layer.

In the code. `gyroresonance/transfer.py`, function `transfer_chunk`: mid-step means (`m(...)`), `dtau_o`, `dtau_x`; the loop `for s in range(1, s_max + 1)` with the test `cross`, the weight `u`, `lerp`, `l_b` (Eq. 33), `layer_optical_depth` for both modes and the $s = 1$ rule; the sums `gr_x`, `gr_o` ($\sum_s \tau_s$) and `st_x`, `st_o` (W). Keyword `s_max`; `mechanisms="gr"` or `"ff"` keeps only one of the two parts.

9 Radiative transfer along the line of sight

For each of the two polarisations p (Sect. 10 says which mode coefficients each one uses in each step) the formal solution of $\mathrm{d}I_p/\mathrm{d}\tau = S_p - I_p$, summed from the observer inwards with S_p constant within a step, is

$$\boxed{I_p = \sum_{k=0}^{N-1} S_k^p \left(1 - e^{-\Delta\tau_k^p}\right) \exp\Big(-\sum_{j<k} \Delta\tau_j^p\Big) \;+\; \frac{\nu^2 k_\mathrm{B}}{c^2}\, T_\mathrm{body}\, e^{-\tau_p}\, \chi_\mathrm{body}} \tag{37}$$

where $\tau_p = \sum_k \Delta\tau_k^p$ is the total optical depth of the line, and the last term (optional) is the emission of the body, with brightness temperature T_body, seen by lines that end on $r = r_\mathrm{body}$ ($\chi_\mathrm{body} = 1$) and are not cut off.

Cutoffs. A polarisation cannot reach the observer from behind a point where its mode does not propagate. If node k is a cutoff node for polarisation p (Eq. 7 evaluated with X_k, Y_k), the steps from k onwards contribute nothing:

$$a_k^p = \prod_{j\le k} \left(1 - c_j^p\right), \qquad \Delta\tau_k^p \to a_{k+1}^p\, \Delta\tau_k^p, \quad W_k^p \to a_{k+1}^p\, W_k^p, \tag{38}$$

where $c_j^p = 1$ at a cutoff node inside the domain and 0 otherwise. Lines that are cut off are flagged (`cutoff`).

In the code. `transfer_chunk`: `alive = cumprod(~(c & live))` (a_k^p), `tau_front = cumsum(d) - d`, `S = sw/d * _SRC * freq**2` (Eq. 36), `w = exp(-tau_front) * -expm1(-d)`, `I = (S*w).sum(axis=1)` (Eq. 37); body term with keyword `t_body` of `make_gr_image` (default: no body emission).

10 Mode coupling and the two polarisations

Along a line of sight $\boldsymbol{B}\cdot\hat{\boldsymbol{k}}$ may change sign (a quasi-transverse region, for instance above the neutral line between two spots). There, the radiation either keeps its mode identity, so that its sense of circular polarisation reverses (*strong coupling*, at low frequencies and strong fields), or keeps its sense of polarisation and changes mode (*weak coupling*, the usual case at microwave frequencies; Cohen 1960). radiosynth offers both limits.

Weak coupling (`coupling="weak"`, default). The transfer is done for right- and left-hand circular polarisation, $p \in \{R, L\}$. In each step

$$(\Delta\tau_k^R, W_k^R) = \begin{cases} (\Delta\tau_k^x, W_k^x), & \overline{\cos\theta}_k > 0, \\ (\Delta\tau_k^o, W_k^o), & \overline{\cos\theta}_k \le 0, \end{cases} \qquad (\Delta\tau_k^L, W_k^L) = \text{the other mode}, \tag{39}$$

and the cutoff of node k is that of the x-mode for R when $\cos\theta_k > 0$ (and of the o-mode otherwise), and conversely for L. Then

$$I = I_R + I_L, \qquad V = I_R - I_L. \tag{40}$$

Strong coupling (`coupling="strong"`). The transfer is done for the modes, $p \in \{x, o\}$, with their own coefficients and cutoffs everywhere, and the sign of V is set by the field direction at the observer's end of the line (the first node with $B > 0$):

$$I = I_x + I_o, \qquad V = \mathrm{sgn}(\cos\theta_{k_1})\,(I_x - I_o)\,. \tag{41}$$

Without a quasi-transverse region the two options give the same result.

In the code. `transfer_chunk`: the `if coupling == "weak"` branch (`pos = cos_m > 0` for the steps, `posn = cos > 0` for the cutoffs) and the `"strong"` branch (`vsign`). Keyword `coupling` of `make_gr_image`.

11 Brightness temperature, polarisation and flux

The image holds, for every frequency and pixel, I and V in $\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$, and the two polarisations I_1, I_2 (R, L or x, o). The Rayleigh–Jeans brightness temperatures are

$$T_b = \frac{c^2}{2k_{\mathrm{B}}\nu^2}\,I, \qquad T_b(V) = \frac{c^2}{2k_{\mathrm{B}}\nu^2}\,V, \qquad T_p = \frac{c^2}{k_{\mathrm{B}}\nu^2}\,I_p \quad (p = 1, 2), \tag{42}$$

so that $T_b = \frac{1}{2}(T_1 + T_2)$ and $T_p = T_e$ for a polarisation that is optically thick in isothermal plasma. The degree of circular polarisation is V/I (undefined where $I = 0$). For a pixel of size $\Delta X \Delta Y$ in the image plane at distance D from the observer, the solid angle is $\Delta\Omega = \Delta X \Delta Y / D^2$ (all in the same unit), and the flux densities are

$$F_I = \sum_{\text{pixels}} I\,\Delta\Omega, \qquad F_V = \sum_{\text{pixels}} V\,\Delta\Omega, \tag{43}$$

in $\mathrm{W\,m^{-2}\,Hz^{-1}}$ ($1\,\mathrm{sfu} = 10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$).

In the code. `gyroresonance/image.py`, class `GRImage`: `intensity`, `stokes_v`, `pol1`, `pol2`, `brightness_temperature`, `tb_v`, `tb_pol`, `polarization`, `flux_density`, `flux_density_v`.

12 Diagnostics

The contribution of step k to the observed intensity of polarisation p is

$$C_k^p = S_k^p \left(1 - e^{-\Delta\tau_k^p}\right) \exp\Big(-\sum_{j<k} \Delta\tau_j^p\Big). \tag{44}$$

It is split between free–free and layers in proportion to their optical depths,

$$C_k^{p,\mathrm{GR}} = C_k^p \,\frac{\sum_s \tau_{s,k}^p}{\Delta\tau_k^p}. \tag{45}$$

From these, for every pixel:

- `gr_fraction` $= \sum_{p,k} C_k^{p,\mathrm{GR}} / \sum_{p,k} C_k^p$, the share of I that comes from the layers;
- `radius` $= \sum_{p,k} C_k^p\,\bar{r}_k / \sum_{p,k} C_k^p$, the intensity-weighted radius of the emission ($\bar{r}_k$: distance of the step's mid-point from the centre);
- `dominant_s`: the harmonic whose layers carry most of the intensity (each step's layer contribution is attributed to the harmonic with the largest τ in that step), 0 if free–free dominates, -1 if there is no emission;
- `x_flag`: true if a significant layer ($\max(\tau^x, \tau^o) > 0.1$) has $X > 0.25$, where the approximation $n_\sigma = 1$ in Eq. (22) becomes rough;
- `cutoff`: true if a polarisation was stopped at a cutoff.

13 Assumptions and range of validity

1. **Maxwellian electrons, lowest order in β_T.** Eq. (22) is the small-Larmor-radius limit; the relative error is $O(s^2\beta_T^2)$, about 1 % or less for $s \le 6$ at coronal temperatures (0.8 % in the test of Sect. 14). Non-thermal electrons are treated by the gyrosynchrotron part of radiosynth.
2. **Refractive index 1.** Valid for $X \ll 1$, normally true where gyroresonance emission forms; layers with $X > 0.25$ are flagged.

3. **o-mode near $\theta = 90°$.** At lowest order $F_o \to 0$ across the field; the true o-mode opacity is then of higher order in β_T and is underestimated.
4. **Straight lines of sight.** Refraction is neglected (it matters only near the plasma level, $X \to 1$, where gyroresonance emission does not form).
5. **Mode coupling** is treated in the two limits, weak or strong, not with the intermediate coupling coefficient.
6. **Magnetised free–free** uses Eq. (27), valid for $Y < 1$ away from quasi-transverse propagation; $Y|\cos\theta|$ is capped at 0.9.
7. **Resolution.** The model must resolve L_B and the variation of n_e and T_e; the step Δs must be small enough that each harmonic is crossed at most once per step.
8. **Chromosphere.** Lines that reach $r_{\rm body}$ see nothing behind it unless a body temperature $T_{\rm body}$ is given.

14 Validation

The test script `tests/test_gyroresonance.py` checks the implementation against independent results:

Test	Result
Layer τ_s vs. Maxwellian average of the exact J^2_{s-1} ($s = 2$–6)	$\leq 0.8\,\%$ ($= O(s^2\beta_T^2)$)
Single $s = 3$ layer in a linear field: $I = S(1 - e^{-\tau})$ per mode	10^{-6}
Thick layers: $T_b(R) = T_b(L) = T_e$	2×10^{-11}
V sign over the positive spot; $\boldsymbol{B} \to -\boldsymbol{B}$ gives $V \to -V$	exact
Straight-ray free–free vs. refractive tracer without refraction	5×10^{-4}
Thin magnetised free–free: $V/I = 2q/(1+q^2)$, Eq. (28)	3×10^{-4}
Spherical grid (B_r, B_θ, B_ϕ) vs. Cartesian image (I; V)	0.8 %; 1.8 %
PLUTO `bx1..3` in code units → gauss	10^{-11}

15 Using it

```
import radiosynth as rs
grid = rs.load_pluto("out/", with_b=True)     # or rs.load_mhd(..., with_b=True)
img = rs.make_gr_image(grid, "3GHz, 5GHz, 17GHz", observer=(-215, 0, 0),
                       x_size=0.4, y_size=0.4, nx=200, ny=200,
                       mechanisms="gr+ff", gaunt="dulk", s_max=10, coupling="weak")
img.plot("tb"); img.plot("pol"); img.plot("harmonic")
print(img.summary())
```

or, from the command line,

```
python -m radiosynth ar.dat --mechanism gyroresonance --freq "5GHz, 17GHz" \
       --size 0.4 0.4 --npix 200 200 --out gr --plot
```

Keyword of `make_gr_image`	Quantity in this note
`mechanisms` (`"gr+ff"`, `"gr"`, `"ff"`)	which of τ_s^σ, $\Delta\tau_{\rm ff}^\sigma$ are included
`gaunt` (`"dulk"` or a number)	$\kappa_{\rm ff}$, Sect. 7
`s_max`	highest harmonic s
`coupling` (`"weak"`, `"strong"`)	Sect. 10
`ds`	step Δs [coordinate units]
`t_body`	$T_{\rm body}$ in Eq. (37)
`observer`, `distance_pc`, `x_size`, `nx`, …	geometry, Sect. 3

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